Cognitive Computation (2025) 17:69
https://doi.org/10.1007/s12559-025-10425-1

RESEARCH

GSAC-UFormer: Groupwise Self-Attention Convolutional Transformer-Based UNet for Medical Image Segmentation

Anass Garbaz[1] · Yassine Oukdach[1] · Said Charfi[1] · Mohamed El Ansari[2] · Lahcen Koutti[1] · Mouna Salihoun[3]

Received: 1 August 2024 / Accepted: 14 February 2025 / Published online: 22 February 2025
© The Author(s), under exclusive licence to Springer Science+Business Media, LLC, part of Springer Nature 2025

Abstract
Traditional transformers struggle to effectively capture local contextual information. Conversely, CNNs face challenges in modeling long-range dependencies. To address these limitations, this paper introduces GSAC-UFormer, an innovative Groupwise Self-Attention Convolutional Transformer-based UNet for medical image segmentation. The design of GSAC-UFormer focuses on efficiently integrating both local and global information, balancing the strengths of different processing techniques. At the core of GSAC-UFormer is the GSAC-Former block. This module combines groupwise convolution with a CNN-adaptive self-attention mechanism, enabling parallel integration of local and global contexts. This architecture allows the model to effectively capture intricate dependencies across various data dimensions while processing local features with high efficiency. The Guided Contextual Feature Attention (GCFA) mechanism further enhances feature selection. It emphasizes the most relevant contextual information, refining spatial and channel-wise relationships in the extracted features. This targeted approach mitigates noise and improves model accuracy. Additionally, the Multi-Depth Partitioned Depthwise Convolution Transformer (MDPDC-Former) serves as a bottleneck module. It optimizes feature mapping and enhances network learning efficiency by dynamically adjusting the receptive field. This enables the model to capture multi-scale semantic information more effectively. Experimental results highlight the superior performance of GSAC-UFormer compared to state-of-the-art methods. It achieves Dice coefficients of 91.6%, 94.61%, and 82.24% on the MICCAI 2017 (red lesion), PH2, and CVC-ClinicalDB datasets, respectively. These results underscore its effectiveness in advancing medical image segmentation.

Keywords GSAC-UFormer · Transformer · UNet · Self-attention · Groupwise convolution · Medical image segmentation

Introduction

Gastrointestinal (GI) bleeding is a frequent issue observed in both hospital emergency rooms and general healthcare settings [1]. Acute or overt bleeding in the GI tract may manifest as hematemesis, melena, or hematochezia. Chronic or occult GI bleeding typically reveals itself through reactive fecal occult blood or iron deficiency anemia, often lacking visible symptoms for the patient. Upper GI bleeding refers to bleeding originating from the esophagus extending to a ligament near the duodenojejunal bend. Lower GI bleeding encompasses bleeding from sites distal to this ligament. Types of lower GI bleeding include colonic bleeding and mid-GI bleeding, which initiates in the small intestine and involves the ampulla of Vater and terminal ileum. Conversely, obscure GI bleeding denotes recurrent bleeding primarily from the small intestine, the origin of which remains unidentified following upper colonoscopy evaluation [2]. Consequently, wireless capsule endoscopy (WCE) [3] has been introduced to investigate small intestine bleeding. The capsule-shaped device, measuring 26 mms in height and 11 mms in width, includes an optical dome, illumination source, video apparatus, power unit, and RF transmitter. It requires ingestion by the patient and transmits between 50,000 and 60,000 visual frames at an average rate of 2 to 4 frames per second over an 8-h period.

Cancers affecting the skin and GI tract are among the most rapidly increasing types worldwide. In 2023, the United States recorded roughly 104,930 new cases of skin cancer and 348,840 GI cancer cases [4]. Skin cancer arises from abnormal cell growth in the outer layer of the skin, typically triggered by DNA damage that remains unrepaired [5]. Among its various forms, melanoma poses the highest risk due to its aggressive nature [6]. It develops from pigment-

Yassine Oukdach, Said Charfi, Mohamed El Ansari, Lahcen Koutti, and Mouna Salihoun contributed equally to this work.

Extended author information available on the last page of the article

producing cells and can spread quickly to distant organs if not addressed early. In GI-related conditions, identifying polyps is essential, as they have the potential to become cancerous. These growths are commonly detected through colonoscopy [2]. Similarly, skin abnormalities are examined using dermoscopy, a key tool in diagnosing suspicious lesions [7].

Various computer-aided systems [8–10] have been developed over time to automate these tasks and assist medical professionals in their examinations. In medical image segmentation, fully convolutional networks (FCNs) [11], notably the U-Net [12], have become popular. Despite their success, CNNs excel at identifying local features but struggle with grasping global context and long-range dependencies. They often fail to integrate features across multiple scales and only capture limited local context in images. For segmentation tasks requiring a blend of fine-grained features and broader contextual understanding, this integration is crucial. On the other hand, transformers [13] and attention mechanisms excel in managing complex dependencies and capturing global context. On the other hand, transformers typically face challenges in effectively capturing local dependencies compared to CNNs.

In this study, we integrate the strengths of CNNs and transformers to create a comprehensive architecture that mitigates their respective limitations. We introduce a novel Groupwise Self-Attention Convolutional Transformer-based UNet for medical image segmentation, named GSAC-UFormer. Initially, a GSAC-Former block is introduced to amalgamate local and global information, integrating a CNN-adaptive self-attention mechanism alongside a novel groupwise convolution. This approach enhances the processing of local features while leveraging self-attention to capture intricate correlations across data dimensions, significantly boosting information extraction capabilities in medical image segmentation. The Guided Contextual Feature Attention (GCFA) method aims to improve feature selection by emphasizing pertinent contextual information. It refines extracted features through significant spatial and channel-wise interactions, ultimately reducing noise in the output. Serving as a bottleneck module linking the encoder and decoder, the Multi-Depth Partitioned Depthwise Convolution Transformer (MDPDC-Former) enhances network learning performance. It facilitates diverse feature transformations by modifying the model's structure and dynamically adjusts the receptive field to capture richer, multi-scale semantic information.

Based on the MICCAI 2017 (red lesion) dataset [14], as well as the PH2 and CVC-ClinicalDB datasets, the experimental results demonstrate that GSAC-UFormer achieves outstanding performance metrics. Specifically, it attains Dice coefficients of 91.6%, 94.61%, and 82.24% on the MICCAI 2017, PH2, and CVC-ClinicalDB datasets, respectively. These results highlight significant advancements and underscore GSAC-UFormer's superiority over state-of-the-art networks.

Related Works

Transformers in Medical Imaging

Transformers' ability to analyze complex spatial and structural data and model long-range dependencies has garnered significant attention in the field of medical imaging [15]. LeViT-UNet, introduced by Xu et al. [16], integrates a LeViT Transformer module into the U-Net architecture. Specifically, they used LeViT for encoding within LeViT-UNet, enhancing the balance between accuracy and efficiency in the transformer block. Additionally, TransUNet was proposed by Chen et al. [17] as a compelling alternative to both U-Net and transformers for medical image segmentation. The transformer component encodes tokenized image patches from a CNN feature map to extract global contexts effectively.

For medical image segmentation, Cao et al. [18] introduced Swin-Unet, a pure transformer analogous to U-Net. This transformer-based U-shaped encoder-decoder design, featuring skip connections, inputs tokenized image patches for local–global semantic feature acquisition. Additionally, Zhang et al. [19] proposed TransFuse, a distinctive parallel-branch architecture. TransFuse effectively captures global dependencies and low-level spatial features in a much shallower manner by combining transformers and CNNs in a parallel configuration. In contrast to pure transformer models and U-Net and its variants, our proposed model integrates self-attention mechanisms and groupwise convolutions concurrently.

Disease Detection in Medical Imaging

Li et al. [20] presented a pixel-level segmentation technique for GI bleeding that uses dual network branches with attention mechanisms to accurately classify pixel samples from bleeding locations. To improve the prediction of lesion types and regions in GI tract endoscopic images, Tang et al. [21] proposed a multi-task network that can learn classification and segmentation simultaneously. This network combines the advantages of CNNs in learning local features with a transformer designed to learn global features. Furthermore, Alshardan et al. [22] introduced an innovative approach called the Harbor Seal Whiskers Optimization Algorithm, integrated with deep learning for medical imaging analysis targeting GI cancer detection. This method aims to enhance the analysis of GI images, facilitating more effective cancer diagnosis.

Ghosh et al. [23] suggested a technique for retrieving local statistical features that considers a block encompassing a

single pixel rather than the value of that single pixel. By integrating the local block properties of three distinct RGB color planes, an index value is defined. These index values yield a color histogram, offering recognizable color texture properties. Furthermore, Amiri et al. [24] proposed a method using the expectation-maximization segmentation methodology to divide the input image into potential zones of interest. Color and quantitative data are then retrieved from these regions. Additionally, a support vector machine is employed to distinguish leaks from normal images.

Recent advancements in polyp detection have introduced various innovative architectures. Zhang et al. [25] developed TranSEFusionNet, which enhances segmentation by utilizing the transformer's ability to capture long-range dependencies. This design emphasizes global contextual understanding, resulting in more accurate detection. Wang et al. [26] proposed MBP-SSNet, an enhanced baseline derived from their earlier MB-PPFG Net framework. Their approach addresses the challenge of encoding and decoding complex features using a multi-bottleneck mechanism, which facilitates the transfer of enriched information. The network also introduces a progressive transmission process that ensures efficient semantic integration.

Li et al. [27] introduced BMCS-Net, a Bi-directional Multi-scale Cascaded Network, designed to improve medical segmentation tasks. The model integrates features from transformer and CNN branches through a unique Two-stream Cascaded Aggregation module, preventing the loss of global semantic details as the network deepens. This integration allows for more precise segmentation by maintaining both local and global feature representations. Feng et al. [28] proposed a hybrid architecture known as the Parallel Contextual Convolutional Transformer. Their model adopts parallel encoder-decoder modules combining contextual transformers and fully convolutional frameworks, ensuring efficient feature extraction. The use of Multi-scale Fusion Output modules and channel-attention skip connections further improves segmentation accuracy by enhancing multi-scale feature integration.

Hasan et al. [29] introduced DSNet, an automated network for precise skin lesion segmentation. To make the model more lightweight, they replaced traditional convolution layers with depthwise separable convolutions, reducing parameter count while projecting discriminative features into pixel space at various encoding stages. Xie et al. [30] proposed the MB-DCNN framework, which simultaneously handles skin lesion segmentation and classification. Their model comprises three interconnected networks: a coarse segmentation module, a mask-guided classification network, and an enhanced segmentation layer to refine results. Ozturk et al. [31] developed iFCN, designed to segment high-resolution skin lesion images without requiring pre- or post-processing steps. Their architecture extends the FCN by incorporating spatial information through a residual structure to improve segmentation accuracy. Alahmadi [32] proposed a semi-supervised learning approach that leverages unlabeled data during training. By incorporating both labeled and unlabeled samples, their method enhances the network's ability to generalize to unseen data.

Liu et al. [33] introduced CSWin-UNet, a U-shaped segmentation architecture that integrates the CSWin self-attention mechanism. This model captures contextual information through horizontal and vertical stripe-based attention, improving feature representation. Wu et al. [34] proposed FAT-Net, a hybrid network combining a CNN and a transformer branch. The transformer branch enhances the model's ability to capture long-range dependencies and global context, which are essential for accurate segmentation. Zhang et al. [35] designed an adaptive position encoding module that improves the fusion of positional information across multiple receptive fields. Their method ensures better token sequence alignment within the transformer architecture, enhancing the model's ability to process spatial details. Heidari et al. [36] presented HiFormer, a cutting-edge model that bridges CNNs and transformers for medical image segmentation. Their design incorporates two multi-scale feature representations, combining a Swin Transformer module with a CNN-based encoder to achieve efficient and accurate segmentation. Unlike the previous studies mentioned, our approach employs both transformer-based and CNN features. This integration enhances segmentation effectiveness by incorporating both local and global information.

Methods

GSAC-UFormer

In this section, we delve deeper into the proposed architecture for segmenting WCE images based on GI bleeding. Figure 1 illustrates the overall structure of the GSAC-UFormer. The proposed architecture comprises three main components: the Groupwise Self-Attention Convolutional Transformer (GSAC-Former), the Guided Contextual Feature Attention (GCFA), and the Multi-Depth Partitioned Depthwise Convolution Transformer (MDPDC-Former).

The GSAC-Former models complex dependencies, capturing both local and global information. To extract high-level contextual and low-level features from the GSAC-Former's output, we introduce the GCFA mechanism. The MDPDC-Former captures both broad, conceptual characteristics and detailed, localized features, enhancing the segmentation process.

The input size is $H \times W \times C$ ($256 \times 256 \times 3$). After each encoding step, the depth dimension increases to $2^n \times C$. This starts with 32 channels. The spatial dimensions are halved,

resulting in $\frac{H}{2^n} \times \frac{W}{2^n}$. n is an integer that increases from 1 to 4. The depth in the MDPDC-Former block doubles at each step, going from the initial depth C to 16C. This process reverses in the decoding path, gradually reducing the depth and restoring the spatial dimensions. We employ GELU as the activation function.

Groupwise Self-attention Convolutional Transformer

The GSAC-Former is depicted in Fig. 1. This transformer enhances feature representation across various receptive fields. It integrates features from different scales and levels. The model extracts contextual information from local to global scales. Groupwise convolution (GWC) and self-attention are applied simultaneously to the data. GWC is particularly effective at detecting a wide range of patterns and variations in the input data. Instead of using standard pointwise convolution, it incorporates depthwise convolution (DW) followed by a multi-channel splitting block. This combination extracts features at multiple scales, capturing both fine details and broader contextual information.

During the DW procedure, a distinctive filter is specified according to (1) for every channel within the input. The t_{th} filter within kernel R is amalgamated with the t_{th} channel in X to generate the t_{th} channel of the resultant feature map X_i. Subsequently, as depicted in Fig. 2, the output is aggregated to yield O_1 (2) and subsequently forwarded through a subsequent multi-channel division block that iterates the process by dispatching it to a group of channels FM with kernel K. The step-by-step mathematical derivation of the computation can be located in (1) and (2).

$$X_i = DW_{r,l,t} = \sum_{i,j} R_{i,j,t} \times X_{r+i-1,l+j-1,t} \tag{1}$$

$$O_1 = Concatenate(\{X_i / i \in \{1, ..., C\}\}) \tag{2}$$

$$O_{1,i} = (FM \times K)_{h,w} = \sum_{j} \sum_{k} K_{j,k} \times FM_{H-i,W-k} \tag{3}$$

$$O_2 = Concatenate(\{GELU(O_{1,i}) / i \in \{1, ..., L\}\}) \tag{4}$$

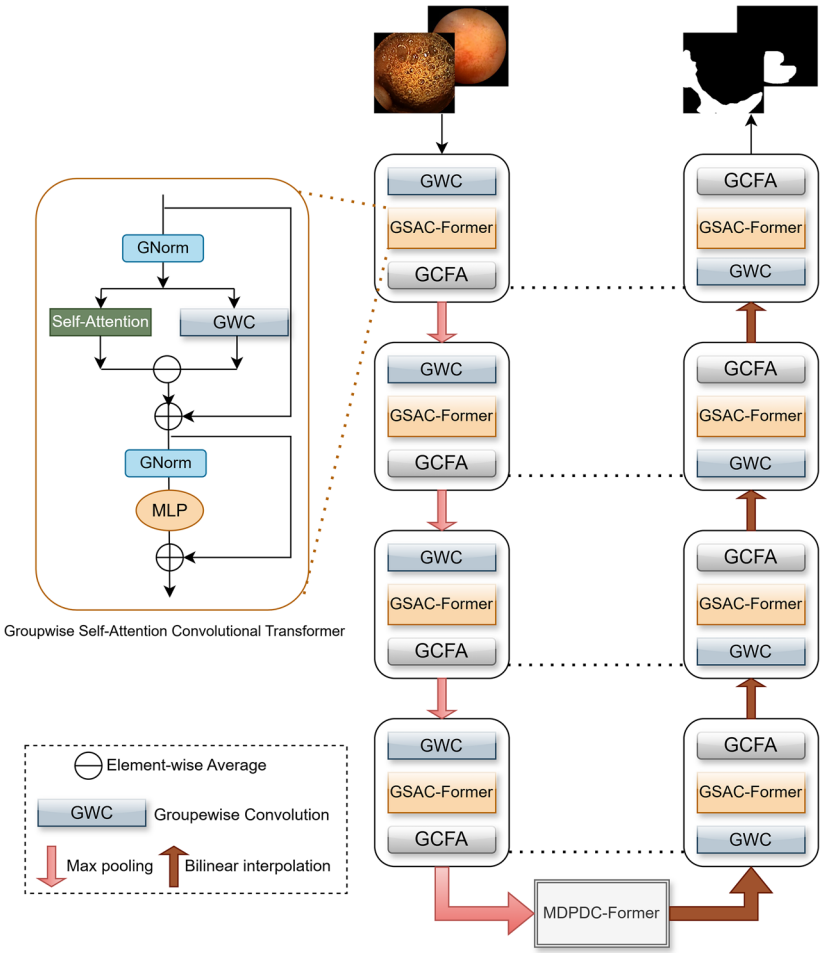


Fig. 1 A visual representation of the proposed GSAC-UFormer design

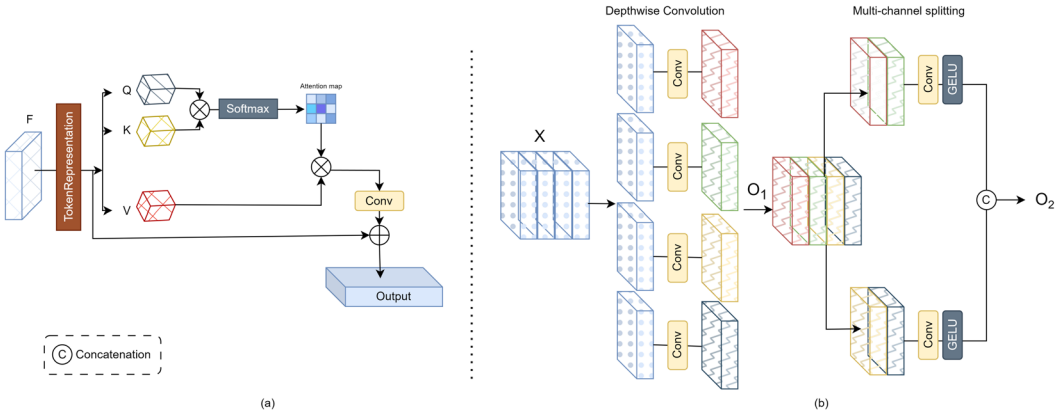


Fig. 2 An overview of GSAC-Former's mechanisms: **a** the adaptive self-attention architecture and **b** the layout of the GWC block

h and w denote the row and column indices of the matrix described in (2). Depending on the depth of the feature map, denoted by L, which varies from block to block, L takes on different values. C represents the number of channels in the input feature map.

The second statement emphasizes that self-attention facilitates a comprehensive understanding of the entire image by effectively capturing global context and long-range dependencies. This capability enables the model to dynamically adjust its focus to different areas of the image as necessary by modifying the receptive field. TokenRepresentations (TR), as defined in (5), serve as the input representation for each token within a sequence. These representations form the basis for the Q (query), K (key), and V (value) vectors.

$$Q = W_q \times \mathrm{TR}(F), \quad K = W_k \times \mathrm{TR}(F), \quad V = W_v \times \mathrm{TR}(F) \tag{5}$$

W_q, W_k, and W_v are the learned weight matrices. We emp loyed the TokenRepresentation module to further decrease the dimensionality of the integrated features. The attention map is then generated as expressed in (6):

$$\mathrm{Attention_Map}(Q, K) = \mathrm{softmax}\left(\frac{Q \cdot K^{\top}}{\sqrt{d_k}}\right) \tag{6}$$

The subsequent step in self-attention is formulated as shown in (7):

$$\mathrm{Output} = \mathrm{Conv}(\mathrm{Attention_Map}(Q, K) \times V) + F \tag{7}$$

Multi-depth Partitioned Depthwise Convolution Transformer

Figure 3 illustrates the proposed MDPDC-Former. It incorporates self-attention, GWC, and DWS within two consecutive

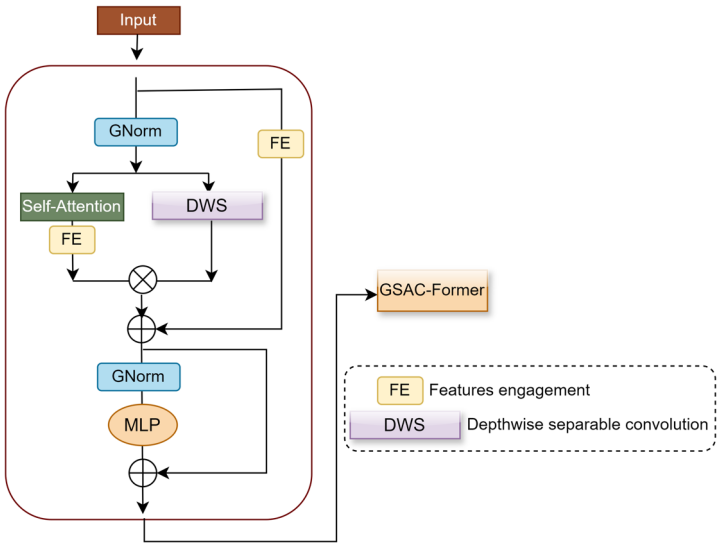


Fig. 3 An illustration of the MDPDC-Former, which serves as the primary system's bottleneck

transformer blocks. This combination enhances segmentation by capturing both high-level, abstract features and fine, localized details. By altering the model's structure and dynamically adjusting the receptive field, it facilitates various feature modifications. This allows the capture of more complex, multi-scale conceptual data.

The input initially passes through a transformer block, beginning with group normalization (GNorm). GNorm mitigates the internal covariate shift. This phenomenon occurs when the distribution of activations shifts during training and can impede learning. The incoming information I follows two parallel paths, akin to GSAC-Former. After a self-attention phase to refine and emphasize the most significant features, FE (convolution operation) takes place. Concurrently, the DWS output undergoes multiplication to eliminate unnecessary information and recover lost data from both sub-blocks. To maintain the feature representation for the decoder, GSAC-Former is positioned after the initial transformers. The subsequent process can be segmented into four primary steps, outlined as follows:

$$Y_1 = \text{FE}\left(\text{Self-Attention}\left(\text{GNorm}(I)\right)\right) \otimes \text{DWS}\left(\text{GNorm}(I)\right) \tag{8}$$

$$Y_2 = \text{FE}(I) + Y_1 \tag{9}$$

$$Y_3 = \text{MLP}\left(\text{GNorm}(Y_2)\right) + Y2 \tag{10}$$

$$Y_4 = \text{GSAC-Former}(Y_3) \tag{11}$$

Guided Contextual Feature Attention Mechanism

After each encoder-decoder transition, the GCFA mechanism (depicted in Fig. 4) is integrated. Its objective is to boost feature selection by prioritizing relevant contextual information. This mechanism leverages robust spatial and channel-wise connections to enhance refined feature extraction, aiming to produce an output with reduced noise.

Initially, $X_{(i,i=1,\ldots,n)}$ is reorganized to $X'_{(i,i=n,1,\ldots,n-1)}$ as defined in (14). This process involves shifting the channels, with the last channel being moved to the front, followed by the remaining channels in their original order. This restructuring aims to enable the network to focus on diverse data characteristics. It potentially improves noise reduction, especially in scenarios that require varied and discriminative feature representations. An adjacent set of GWC and DWS subsequently traverses through $X'_{(i,i=n,1,\ldots,n-1)}$ before reaching the channel attention block (CA). CA diminishes less pertinent feature channels while accentuating selectively crucial ones.

In (15), the Avg$(\cdot)$ operation denotes an average layer designed to compute the element-wise average between the outputs of the two CA branches. Specifically, for each pair of corresponding pixels from the outputs of CA(GWC(X')) and CA(DWS(X')), the layer calculates their average. This operation integrates information from both branches, producing a globally aggregated feature representation.

Following averaging, the results are directed to spatial attention (SA). SA's role is to concentrate on specific spatial regions within the averaged feature map. The subsequent sections detail the mathematical formulations for spatial and channel attention:

$$SA(FM) = \hat{FM}_{i,j,c} = \sigma(k^{7x7} \times ([FM_{avg}; FM_{max}])) \otimes FM_{i,j,c} \tag{12}$$

$$CA(FM) = \hat{FM}_{i,j,c} = \sigma(W_1.ReLU(\frac{W_0}{H \times W} \sum_{i}^{H} \sum_{j}^{W} X_{i,j,c})) \otimes FM_{i,j,c} \tag{13}$$

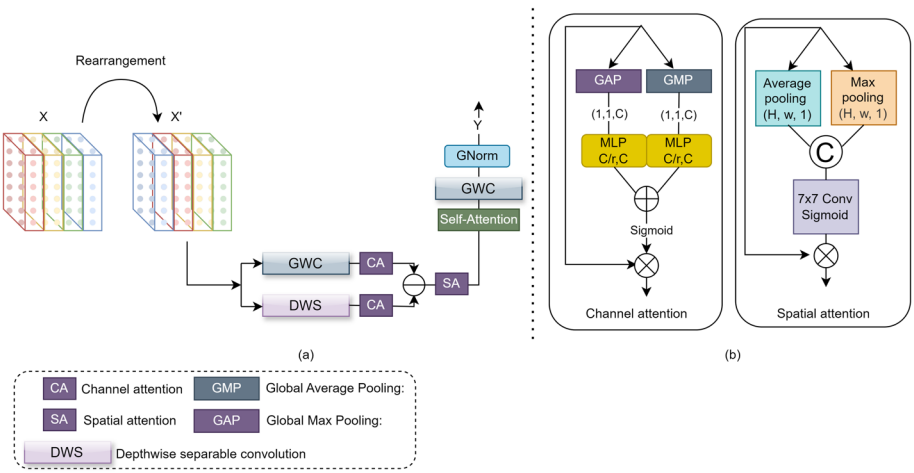


Fig. 4 Illustration of the GCFA process: **a** the primary design of the GCFA and **b** an example of the GCFA block's use of spatial-channel attention

Given that $r = 8$ denotes the reduction ratio and σ represents the sigmoid function. The weight matrices are defined as $W_0 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $W_1 \in \mathbb{R}^{C \times \frac{C}{r}}$. The max-pooled feature map is denoted as $F_{\max}$, while the channel-wise average is represented by F_{avg}. These two features are concatenated to form $[F_{\text{avg}}; F_{\max}]$.

Convolution with a 7×7 filter is indicated by $k^{7\times7}$. To maintain consistency, incorporating self-attention, GWC, and GNorm in the final phase helps prevent confusion with different representations in the following receiving block.

The core steps of the computation are repeated in the following equations:

$$X'_{(i,i=n,1,\ldots,n-1)} = \text{Rearrangement}(X_{(i,i=1,\ldots,n)}) \tag{14}$$

$$X'_1 = \text{SA}\left(\text{Avg}\left(\text{CA}\left(\text{GWC}(X')\right)\right), \text{CA}\left(\text{DWS}(X')\right)\right) \tag{15}$$

$$Y = \text{GNorm}\left(\text{GWC}\left(\text{Self-Attention}(X'_1)\right)\right) \tag{16}$$

Experiments

Datasets

The dataset [14] presented at the MICCAI 2017 Endoscopic Vision Sub-Challenge included 3895 WCE frames, each with a resolution of 320×320 pixels. This dataset comprised frames depicting both normal tissue and various lesions, such as hemorrhages and angioectasias. In this case, the normal class, which consists solely of black masks, is excluded. The dataset was split into two portions: 20% was allocated for testing and validation, while the remaining 80% was used for training.

The second dataset, CVC-ClinicalDB [37], contains a collection of 612 images derived from 31 colonoscopy video sequences. Each image has a resolution of 384×288 pixels. The dataset is specifically curated to support the task of detecting polyps in colonoscopy footage. It is divided into two subsets: 82% for training and 18% for testing. Additionally, the training set is expanded to include 2000 augmented images created using techniques such as rotations, flips, and variations in brightness.

The third dataset, PH2 [38], comprises a total of 200 images, predominantly showcasing cases of naevus and melanoma. These images are formatted in 8-bit RGB with a resolution of 768×560 pixels and were captured using a lens with $20\times$ magnification. The dataset is partitioned into 80% for training purposes and 20% for testing.

Implementation Details

We implemented the TensorFlow-based GSAC-UFormer model using an NVIDIA RTX 3090 24GB DDR6X graphics card. Training was conducted with a batch size of 2 and was limited to 150 epochs.

The Adam optimizer was employed with a learning rate (α) of 0.001. The values for β_1 and β_2 were set to 0.8 and 0.99, respectively. Input images were cropped to a size of 256×256. The number of channels $N \in \{32, 64, 128, 256, 512\}$ was used for the four encoder-decoder blocks and the bottleneck. An overview of this design configuration is provided in Table 1.

Loss Function and Evaluation Metrics

We explored WCE image segmentation using two commonly adopted metrics in scientific studies. The first metric is intersection over union (IoU), which evaluates the overlap between the target mask (G) and the predicted segmentation result (S).

The second metric is the Dice coefficient (DC), which considers both precision and recall when assessing the agreement between the predicted segmentation and the ground truth. DC is a reliable metric for evaluating segmentation accuracy, as it is sensitive to changes in segmentation. The following formulas are used to determine the DC and IoU:

$$IoU = \frac{|G \cap S|}{|G| \cup |S|} \tag{17}$$

$$DC = \frac{2 \times |S \cap G|}{|S| + |G|} \tag{18}$$

The loss function we have selected comprises two components: binary cross entropy (BCE) and Dice loss. The following formula is used to calculate the Dice loss (L_{dice}):

$$L_{Dice} = \frac{2 \times \sum Y_{true} \times Y_{pred}}{\sum Y^2_{true} + \sum Y^2_{pred}} \tag{19}$$

Table 1 Configuration parameters for the proposed GSAC-UFormer

Configuration	Value
Optimizer	Adam
Learning rate	0.001
β_1/β_2	0.8/0.99
Epochs	150
Batch size	2
GPU	NVIDIA RTX 3090 24 G DDR6X
Channels	{32, 64, 128, 256, 512}

Y_{true} and Y_{pred} represent the ground truth and predicted segmentation masks, respectively, in the Dice loss calculation. The final loss function is as follows:

$$L_{loss} = L_{bce} + L_{dice} \tag{20}$$

Results

To evaluate the performance of GSAC-UFormer, we conducted a comprehensive comparative study. The reviewed models included U-Net [12], U-Net++ [39], Attention U-Net [40], TernausNet [41], ResUNet++ [42], ResUNet-a [43], V-Net [44], and SegNet [45], along with CNN-transformer-based methods and other state-of-the-art techniques. To ensure a fair comparison, the standard models included in our evaluation were implemented with identical parameter settings to those used for our proposed approach, despite being trained on different datasets. This alignment guarantees that any observed performance differences are attributable to the underlying model architectures, rather than disparities in training configurations.

Results on MICCAI 2017 Dataset

Table 2 summarizes the results of various segmentation techniques on the MICCAI 2017 red lesion dataset. The performance of our proposed GSAC-UFormer is noticeably superior to that of the majority of state-of-the-art models. GSAC-UFormer's remarkable performance in segmentation tasks can be attributed to its comprehensive methodology. It excels in handling complex dependencies, capturing global context, and efficiently extracting local features through structured learning.

U-Net [12] and SegNet [45], having similar basic structures, produce comparable results. Attention U-Net [40], Ternausnet [41], and V-Net [44] demonstrate competitive performance across several evaluation criteria. GSAC-UFormer outperforms ResUNet-a [43], the model ranked second, by 0.07%. Compared to ResUNet++ [42], GSAC-UFormer shows a 0.71% improvement in IoU and a 0.42% gain in DC.

By comparing the computational features and parameter counts of different segmentation models, Table 2 provides a comprehensive evaluation. Notably, GSAC-UFormer demonstrates higher computational efficiency with lower memory usage and fewer parameters.

Moreover, GSAC-UFormer outperformed the transformer-based model, TransCS-Net [46]. It achieved a 0.49% higher DC, demonstrating superior segmentation performance. This improvement was accompanied by a significant reduction in computational complexity. GSAC-UFormer required 56.94 fewer GFLOPs compared to TransCS-Net. This result highlights GSAC-UFormer's remarkable efficiency in handling segmentation tasks. It delivers competitive accuracy while maintaining much lower computational demands. These qualities make it an excellent choice for resource-constrained environments.

Results on CVC-ClinicalDB Dataset

Regarding polyp segmentation, we evaluated GSAC-UFormer on the CVC-ClinicalDB dataset [37], as shown in Table 3. The proposed method outperformed standard models, showcasing its effectiveness in segmentation tasks. Compared to transformer-based approaches, the best-performing model, PCCTrans [28], achieved a higher DC by a margin of only 5.94%. However, GSAC-UFormer demonstrated significantly lower computational complexity, requiring just 1.39 GFLOPs compared to PCCTrans's 38.5 GFLOPs. This emphasizes GSAC-UFormer's remarkable efficiency.

Additionally, GSAC-UFormer demonstrated superior computational efficiency when compared to other advanced

Table 2 Evaluation of GSAC-UFormer compared to advanced segmentation techniques using the MICCAI 2017 dataset

Method	DC (%)	IoU (%)	GFLOPs	Params (M)
SegNet [45]	72.37	56.71	51.04	11.74
U-Net [12]	72.37	56.71	101.75	34.51
TransUNet [17]	72.37	56.71	89.4	17.39
U-Net++ [39]	80.45	67.29	59.6	9.04
Attention U-Net [40]	90.16	82.09	135.14	35.23
V-Net [44]	90.54	82.72	21.76	23.75
TernausNet [41]	90.67	82.93	62.33	23.01
TransCS-Net [46]	91.11	**85.52**	58.33	51.97
ResUNet++ [42]	91.18	83.80	32.07	4.07
MLFA-UNet [17]	91.42	84.2	36.95	7.91
ResUNet-a [43]	91.53	84.38	30.04	6.28
GSAC-UFormer	**91.6**	84.51	**1.39**	**3.3**

For each column, the best results are marked in bold

Table 3 Evaluation of the effectiveness of different approaches on the CVC-ClinicalDB dataset

Method	DC (%)	IoU (%)	GFLOPs	Params (*M*)
ResUNet++ [42]	79.55	79.62	32.07	4.07
V-Net [44]	79.59	66.10	21.76	23.75
U-Net [12]	81.10	68.21	101.7	534.51
TranSEFusionNet [25]	86.48	79.09	124.43	127.74
MBP-SSNet [26]	86.57	78.24	105.6	9.37
BMCS-Net [27]	88	**81.7**	35.33	27.03
PCCTrans [28]	**88.18**	–	38.5	50.8
GSAC-UFormer	82.24	69.85	**1.39**	**3.3**

An en dash "–" indicates missing data

transformer models. For example, BMCS-Net [27] requires 35.33 GFLOPs. These results highlight GSAC-UFormer's ability to balance high segmentation accuracy with minimal computational demands. This makes it a practical choice for real-world applications.

Results on PH2 Dataset

For skin segmentation, we evaluated the proposed GSAC-UFormer on the PH2 dataset [38], as shown in Table 4. Among CNN-based methods, the top-performing model, iFCN [31], achieved an IoU of 87.6%. Its computational complexity was 72 GFLOPs, and it had 49 million parameters. GSAC-UFormer exceeded iFCN's performance by a margin of 2.18% in IoU. It also significantly reduced the GFLOPs by 70.61. This demonstrates GSAC-UFormer's superior accuracy and computational efficiency.

When compared to transformer-based models, GSAC-UFormer surpassed CSWin-UNet [33] in terms of computational efficiency. The difference in DC was marginal at 0.32%. The best-performing transformer model, HiFormer [36], achieved a DC of 94.6%. This performance closely aligns with GSAC-UFormer. However, GSAC-UFormer maintained a substantial advantage in parameter count and computational complexity. These results highlight its efficiency and suitability for real-world applications.

Ablation Study

Ablation Study of Subnetworks

We assessed the effects of several modules included in GSAC-UFormer, specifically GCFA and MDPDC-Former, as shown in Table 5. In GSAC-Former, we evaluated the impact of utilizing GWC and Self-attention independently. These modules were used to capture both local and global dependencies.

To determine how the GCFA module might enhance feature selection by prioritizing relevant contextual information, we first integrated it into model D. This integration resulted in a DC score of 91.36% and an IoU of 84.09%. We then applied the same technique to the MDPDC-Former in model C. This also yielded favorable outcomes, with DC and IoU reaching 91.27% and 83.94%, respectively. The module's ability to capture both fine-grained, regional information and high-level, conceptual characteristics for the decoder may impact this performance. Next, we combined GCFA and GSAC-Former with GWC alone in model A. Compared to models D and C, it performed poorly. The direct connection between GCFA and the GSAC-Former, where GCFA receives distinct information flow and subsequently transmits it to the next block, may partially account for this low DC. As a result, it incorrectly considers noisy areas to be important. In model

Table 4 Comparison of GSAC-UFormer with advanced segmentation techniques using the PH2 dataset

Model	DC (%)	IoU (%)	GFLOPs	Params (*M*)
DSNet [29]	92.00	87.2	35.0	10.0
MB-DCNN [30]	93.30	87.1	55.0	53.0
iFCN [31]	93.20	87.6	72.0	49.0
Alahmadi et el. [32]	94.01	–	–	–
CSWin-UNet [33]	94.29	–	4.72	23.57
FAT-Net [34]	94.40	89.6	23.0	30.0
APT-Net [35]	94.60	**89.9**	22.47	–
HiFormer [36]	94.60	–	8.05	25.51
GSAC-UFormer	**94.61**	89.78	**1.39**	**3.3**

Table 5 The influences of various modules and mechanisms are examined through an ablation study using the MICCAI 2017 dataset

No.	GCFA	MDPDC-Former	Self-attention	GWC	DC (%)	IoU (%)
A	✓			✓	90.37	82.44
B		✓		✓	91.25	83.92
C		✓			91.27	83.94
D	✓				91.36	84.09
E	✓		✓		91.6	84.5
F	✓	✓	✓		**91.6**	**84.51**

B, the same procedure is used for MDPDC-Former, resulting in a DC of 91.25%.

The optimal outcome was then obtained by repeating the process with the self-attention module in model E, which is related to the transmission of identical contextual characteristics. Ultimately, model E achieves the best results by integrating the entire system and combining all of its benefits.

The modest improvement of 0.01 in IoU upon adding the MDPDC-Former module in model F can be attributed to several factors. First, the integration of GCFA and self-attention modules in model E has already optimized feature representations, leading to a saturation of performance gains. Second, the MDPDC-Former's role in dynamically handling multi-scale feature representations enhances fine-grained details but contributes only localized improvements, which have minimal impact on aggregate metrics like IoU. Third, overlapping functionalities between MDPDC-Former, GCFA, and self-attention modules result in diminishing returns when combined. Finally, the MDPDC-Former primarily enhances the model's robustness in handling challenging edge cases and ambiguous regions, benefits that are critical for practical robustness and deployment but may not significantly influence overall IoU gains.

Additionally, as illustrated in Fig. 5, we offer a thorough comparison analysis of the receiver operating characteristic (ROC) curves and the corresponding area under the curve (AUC) values. This graphic demonstrates the consistent improvements our model achieves with different module combinations, explaining the previously reported results.

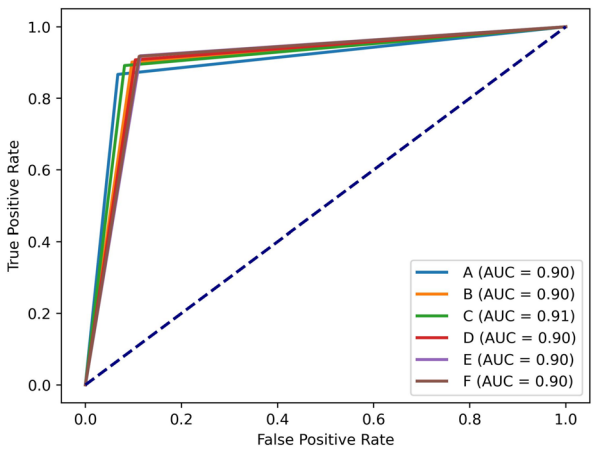


Fig. 5 ROC curves showing the performance variations of GSAC-UFormer across different module and mechanism configurations

Ablation Study of Hyperparameters

A thorough description of the ablation study's conclusions is given in Table 6. We carefully tested several hyperparameter combinations to optimize the performance of GSAC-UFormer. The Adam optimizer's default values for the α, β_1, and β_2 hyperparameters are 0.001, 0.99, and 0.999, respectively. In this basic setup, model D achieves 91.48% in DC. Next, we examined the consequences of changing the β values while keeping α constant. Model B experienced a drop in DC. Following this, we attempted to lower the α value, but model A's DC dropped even further, to 89.02%.

Model C achieved a DC of 91.6% by lowering β values while maintaining α at the default level. These findings demonstrate how the model's performance is sensitive to variations in the β parameters. We also present a comprehensive comparison of ROC curves and corresponding AUC values in Fig. 6. Across an array of hyperparameter combinations, our model consistently outperforms the competition.

Discussion

This work addresses the challenging problem of segmenting WCE frames. The primary difficulties arise from the shifting bleeding regions and the confusing surrounding features. The results from the previous section demonstrate that the pro-

Table 6 Exploration of various hyperparameter combinations through an ablation study using the MICCAI 2017 dataset

No.	α	$\beta 1$	$\beta 2$	DC (%)	IoU (%)
A	0.00001	0.99	0.999	89.02	80.21
B	0.0001	0.99	0.999	90.74	83.05
C	0.0001	0.89	0.99	91.22	83.86
D	0.001	0.99	0.999	91.48	84.3
E	0.001	0.8	0.99	**91.6**	**84.5**

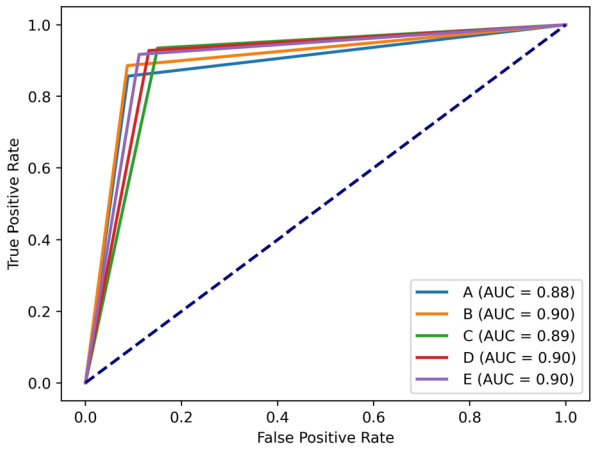


Fig. 6 ROC curves demonstrating GSAC-UFormer's performance across different hyperparameter configurations

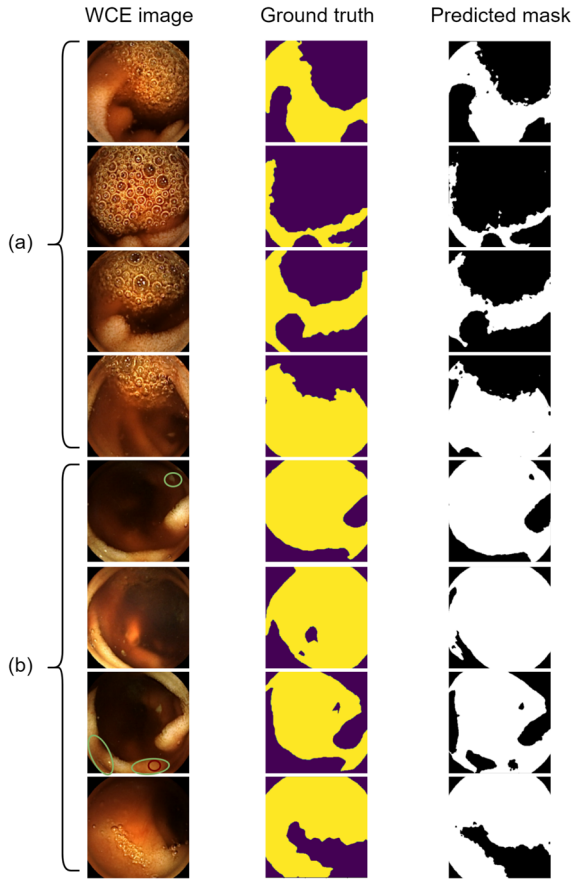


Fig. 7 Display of the segmentation results obtained with GSAC-UFormer on the MICCAI 2017 dataset: **a** various instances segmented into regions and **b** detailed segmentations with highly precise detection

posed strategy outperforms various current methods on the MICCAI 2017 dataset. The proposed approach is applied to a range of diseases. It is effective across different imaging modalities. These include polyp detection in colonoscopy images and skin lesion analysis using dermoscopy.

Moreover, CNNs, while excellent at detecting regional contextual information between nearby pixels or patches, may struggle to understand long-distance connections in images. On the other hand, transformers excel at identifying long-distance dependencies in sequential data. However, the transformer's construction does not explicitly provide location information.

In light of this, this study introduces GSAC-UFormer, a novel Groupwise Self-Attention Convolutional Transformer-based U-Net designed for medical image segmentation. GSAC-UFormer combines the advantages of both transformers and CNNs.

Despite GSAC-UFormer's promising segmentation performance, certain limitations persist. For example, as shown in Fig. 7b in the third line, where we explored two green zones with some bleeding that appears to be normal, the mask incorrectly classifies these areas as normal. These limitations may be related to the module's competencies. Additionally, we examined a red zone that appeared unaffected by bleeding and was classified as normal by the model in the second green zone. Similarly to the first row in Fig. 7b, the small green zone contained an anomalous green object that the model misclassified as normal.

In Fig. 7a, the first two examples were successful in segmenting bleeding zones, although a few edges produced noisy data that did not follow a strict arc. The complexity of feature development needed to distinguish minute details may be contributing to this issue. In future work, we aim to refine GSAC-UFormer by reducing the interconnected modules responsible for creating contextual features. This approach is intended to produce a more streamlined version of GSAC-UFormer.

Conclusion

The GI system can exhibit a range of abnormalities, such as cancers, polyps, and inflammatory bowel disease. GI bleeding can also signal specific conditions. Given that precise lesion localization is crucial, there is a strong push among healthcare professionals for advanced methods in bleeding segmentation. In this work, we leverage the strengths of both CNNs and transformers to create a comprehensive architecture that mitigates the limitations of each. We introduce GSAC-UFormer, a novel U-Net designed for medical image segmentation, based on a Groupwise Self-Attention Convolutional Transformer. The trials presented here show that the proposed method outperforms several state-of-the-art frameworks. The effectiveness of each component of our methodology is further validated by the ablation study results. However, a significant issue remains: even if a noisy region is not included in the mask, the model may still classify it

as important. This can lead to increased edge noise between bleeding and normal areas. In future work, we aim to enhance GSAC-UFormer by reducing the number of interconnected modules that provide contextual data. This approach aims to develop a more efficient version of GSAC-UFormer.

Author Contribution A.G., Y.O, S.C., M.E.A., L.K. and M.S. participated in the conception and design of the study, performed the statistical analysis and investigations, validated the results, and drafted the manuscript.

Funding This work was supported by the Ministry of National Education by Vocational Training; in part by the Higher Education and Scientific Research through the Ministry of Industry, Trade, and Green and Digital Economy; in part by the Digital Development Agency (ADD); and in part by the National Center for Scientific and Technical Research (CNRST) under Project ALKHAWARIZMI/2020/20.

Data Availability The datasets analysed during the current study are available in the Red Lesion Endoscopy repository, [https://rdm.inesctec.pt/dataset/nis-2018-003], in the PH2 database repository, [https://www.fc.up.pt/addi/ph2%20database.html], and in the CVC-ClinicDB repository, [https://www.kaggle.com/datasets/balraj98/cvcclinicdb]

Declarations

Consent to Participate All participants in the study provided written informed consent.

Consent for Publication The manuscript does not include any identifiable individual data or images that would necessitate obtaining consent to publish from any participant.

Conflict of Interest The authors declare no competing interests.

References

1. Kim BSM, Li BT, Engel A, Samra JS, Clarke S, Norton ID, Li AE. Diagnosis of gastrointestinal bleeding: a practical guide for clinicians. World J Gastrointest Pathophysiol. 2014;5(4):467.
2. Fisher DA, Maple JT, Ben-Menachem T, Cash BD, Decker GA, Early DS, Evans JA, Fanelli RD, Fukami N, Hwang JH, et al. Complications of colonoscopy. Gastrointest Endosc. 2011;74(4):745–52.
3. Iddan G, Meron G, Glukhovsky A, Swain P. Wireless capsule endoscopy. Nature. 2000;405(6785):417–417.
4. Siegel RL, Miller KD, Wagle NS, Jemal A. Cancer statistics, 2023. CA: a cancer journal for clinicians. 2023;73(1).
5. Linares MA, Zakaria A, Nizran P. Skin cancer. Prim Care. 2015;42(4):645–59.
6. Schadendorf D, Van Akkooi AC, Berking C, Griewank KG, Gutzmer R, Hauschild A, Stang A, Roesch A, Ugurel S. Melanoma. Lancet. 2018;392(10151):971–84.
7. Ring C, Cox N, Lee JB. Dermatoscopy. Clin Dermatol. 2021;39(4):635–42.
8. Lafraxo S, Souaidi M, El Ansari M, Koutti L. Semantic segmentation of digestive abnormalities from WCE images by using AttResU-Net architecture. Life. 2023;13(3):719.
9. Musha A, Hasnat R, Mamun AA, Ping EP, Ghosh T. Computer-aided bleeding detection algorithms for capsule endoscopy: a systematic review. Sensors. 2023;23(16):7170.
10. Nemani P, Vadali VSS, Medi PR, Marisetty A, Vollala S, Kumar S. Cross-modal hybrid architectures for gastrointestinal tract image analysis: a systematic review and futuristic applications. Image Vis Comput. 2024;105068.
11. Long J, Shelhamer E, Darrell T. Fully convolutional networks for semantic segmentation. In: Proceedings of the IEEE conference on computer vision and pattern recognition. 2015. p. 3431–3440.
12. Ronneberger O, Fischer P, Brox T. U-Net: convolutional networks for biomedical image segmentation. In: Medical image computing and computer-assisted intervention–MICCAI 2015: 18th International Conference, Munich, Germany, October 5-9, 2015, Proceedings, Part III 18. Springer; 2015. p. 234–241.
13. Vaswani A, Shazeer N, Parmar N, Uszkoreit J, Jones L, Gomez AN, Kaiser Ł, Polosukhin I. Attention is all you need. Adv Neural Inf Process Syst. 2017;30.
14. Coelho P, Pereira A, Leite A, Salgado M, Cunha A. A deep learning approach for red lesions detection in video capsule endoscopies. In: Image analysis and recognition: 15th International Conference, ICIAR 2018, Póvoa de Varzim, Portugal, June 27–29, 2018, Proceedings 15. Springer; 2018. p. 553–561.
15. Shamshad F, Khan S, Zamir SW, Khan MH, Hayat M, Khan FS, Fu H. Transformers in medical imaging: a survey. Med Image Anal. 2023;88:102802.
16. Xu G, Zhang X, He X, Wu X. LeViT-UNet: make faster encoders with transformer for medical image segmentation. In: Chinese Conference on Pattern Recognition and Computer Vision (PRCV). Springer; 2023. p. 42–53.
17. Chen J, Lu Y, Yu Q, Luo X, Adeli E, Wang Y, Lu L, Yuille AL, Zhou Y. TransUNet: transformers make strong encoders for medical image segmentation. 2021. arXiv:2102.04306.
18. Cao H, Wang Y, Chen J, Jiang D, Zhang X, Tian Q, Wang M. Swin-UNet: UNet-like pure transformer for medical image segmentation. In: European conference on computer vision. Springer; 2022. p. 205–218.
19. Zhang Y, Liu H, Hu Q. Transfuse: fusing transformers and CNNs for medical image segmentation. In: Medical image computing and computer assisted intervention–MICCAI 2021: 24th International Conference, Strasbourg, France, September 27–October 1, 2021, Proceedings, Part I 24. Springer; 2021. p. 14–24.
20. Li S, Si P, Zhang Z, Zhu J, He X, Zhang N. DFCA-Net: dual feature context aggregation network for bleeding areas segmentation in wireless capsule endoscopy images. J Med Biol Eng. 2022;42(2):179–88.
21. Tang S, Yu X, Cheang CF, Liang Y, Zhao P, Yu HH, Choi IC. Transformer-based multi-task learning for classification and segmentation of gastrointestinal tract endoscopic images. Comput Biol Med. 2023;157:106723.
22. Alshardan A, Saeed MK, Alotaibi SD, Alashjaee AM, Salih N, Marzouk R. Harbor seal whiskers optimization algorithm with deep learning-based medical imaging analysis for gastrointestinal cancer detection. Health Inf Sci Syst. 2024;12(1):1–13.
23. Ghosh T, Fattah SA, Wahid KA. CHOBS: color histogram of block statistics for automatic bleeding detection in wireless capsule endoscopy video. IEEE J Transl Eng Health Med. 2018;6:1–12.
24. Amiri Z, Hassanpour H, Beghdadi A. A computer-aided method to detect bleeding frames in capsule endoscopy images. In: 2019 8th European workshop on visual information processing (EUVIP). IEEE; 2019. p. 217–221.
25. Zhang Y, Liu L, Han Z, Meng F, Zhang Y, Zhao Y. TranSEFusionNet: deep fusion network for colorectal polyp segmentation. Biomed Signal Process Control. 2023;86:105133.
26. Wang Y, Zhang Y, Zhang L, Xu Y, Feng R, Cai H, Xue J, Zhao Z, Guo X, Wei Y, et al. Multi-bottleneck progressive propulsion network for medical image semantic segmentation with integrated macro-micro dual-stage feature enhancement and refinement. Expert Syst Appl. 2024;252:124179.

27. Li B, Wang J, Wang B, Shao Z, Li W, Huang J, Li P. BMCS-Net: a bi-directional multi-scale cascaded segmentation network based on transformer-guided feature aggregation for medical images. Comput Biol Med. 2024;180:108939.
28. Feng Y, Su J, Zheng J, Zheng Y, Zhang X. A parallelly contextual convolutional transformer for medical image segmentation. Biomed Signal Process Control. 2024;98:106674.
29. Hasan MK, Dahal L, Samarakoon PN, Tushar FI, Martí R. DSNet: automatic dermoscopic skin lesion segmentation. Comput Biol Med. 2020;120:103738.
30. Xie Y, Zhang J, Xia Y, Shen C. A mutual bootstrapping model for automated skin lesion segmentation and classification. IEEE Trans Med Imaging. 2020;39(7):2482–93.
31. Öztürk Ş, Özkaya U. Skin lesion segmentation with improved convolutional neural network. J Digit Imaging. 2020;33:958–70.
32. Alahmadi MD, Alghamdi W. Semi-supervised skin lesion segmentation with coupling CNN and transformer features. IEEE Access. 2022;10:122560–9.
33. Liu X, Gao P, Yu T, Wang F, Yuan R-Y. CSWin-UNet: transformer UNet with cross-shaped windows for medical image segmentation. Inf Fusion. 2025;113:102634.
34. Wu H, Chen S, Chen G, Wang W, Lei B, Wen Z. FAT-Net: feature adaptive transformers for automated skin lesion segmentation. Med Image Anal. 2022;76:102327.
35. Zhang N, Yu L, Zhang D, Wu W, Tian S, Kang X. APT-Net: adaptive encoding and parallel decoding transformer for medical image segmentation. Comput Biol Med. 2022;151:106292.
36. Heidari M, Kazerouni A, Soltany M, Azad R, Aghdam EK, Cohen-Adad J, Merhof D. HiFormer: hierarchical multi-scale representations using transformers for medical image segmentation. In: Proceedings of the IEEE/CVF winter conference on applications of computer vision. 2023. p. 6202–6212.
37. Bernal J, Sánchez FJ, Fernández-Esparrach G, Gil D, Rodríguez C, Vilariño F. WM-DOVA maps for accurate polyp highlighting in colonoscopy: validation vs. saliency maps from physicians. Comput Med Imaging Graph. 2015;43:99–111.
38. Mendonça T, Ferreira PM, Marques JS, Marcal AR, Rozeira J. Ph 2-a dermoscopic image database for research and benchmarking. In: 2013 35th annual international conference of the IEEE engineering in medicine and biology society (EMBC). IEEE; 2013. p. 5437–5440.
39. Zhou Z, Siddiquee MMR, Tajbakhsh N, Liang J. UNet++: redesigning skip connections to exploit multiscale features in image segmentation. IEEE Trans Med Imaging. 2019;39(6):1856–67.
40. Oktay O, Schlemper J, Folgoc LL, Lee M, Heinrich M, Misawa K, Mori K, McDonagh S., Hammerla, N.Y., Kainz, B., et al.: Attention U-Net: learning where to look for the pancreas. 2018. arXiv:1804.03999.
41. Iglovikov V, Shvets A. TernausNet: U-Net with VGG11 encoder pre-trained on ImageNet for image segmentation. 2018. arXiv:1801.05746.
42. Jha D, Smedsrud PH, Riegler MA, Johansen D, De Lange T, Halvorsen P, Johansen HD. ResUNet++: an advanced architecture for medical image segmentation. In: 2019 IEEE International symposium on multimedia (ISM). IEEE; 2019. p. 225–2255.
43. Diakogiannis FI, Waldner F, Caccetta P, Wu C. ResUNet-a: a deep learning framework for semantic segmentation of remotely sensed data. ISPRS J Photogramm Remote Sens. 2020;162:94–114.
44. Milletari F, Navab N, Ahmadi S-A. V-Net: fully convolutional neural networks for volumetric medical image segmentation. In: 2016 Fourth international conference on 3d vision (3DV). IEEE; 2016. p. 565–571.
45. Badrinarayanan V, Kendall A, Cipolla R. SegNet: a deep convolutional encoder-decoder architecture for image segmentation. IEEE Trans Pattern Anal Mach Intell. 2017;39(12):2481–95.
46. Tang S, Cheang CF, Yu X, Liang Y, Feng Q, Chen Z. Transcs-Net: a hybrid transformer-based privacy-protecting network using compressed sensing for medical image segmentation. Biomed Signal Process Control. 2023;86:105131.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.

Springer Nature or its licensor (e.g. a society or other partner) holds exclusive rights to this article under a publishing agreement with the author(s) or other rightsholder(s); author self-archiving of the accepted manuscript version of this article is solely governed by the terms of such publishing agreement and applicable law.

Authors and Affiliations

Anass Garbaz[1] · Yassine Oukdach[1] · Said Charfi[1] · Mohamed El Ansari[2] · Lahcen Koutti[1] · Mouna Salihoun[3]

✉ Anass Garbaz
anass.garbaz@gmail.com

Yassine Oukdach
ysoukdach@gmail.com

Said Charfi
charfisaid@gmail.com

Mohamed El Ansari
moh.elansari@umi.ac.ma

Lahcen Koutti
l.koutti@uiz.ac.ma

Mouna Salihoun
mounasalihoun@gmail.com

[1] LabSIV, Department of Computer Science Faculty of Sciences, Ibn Zohr University, Agadir, Morocco

[2] Informatics and Applications Laboratory, Department of Computer Science Faculty of Sciences, My Ismail University, Meknes, Morocco

[3] Faculty of Medicine and Pharmacy, Mohammed V University, 10100 Rabat, Morocco